INCH-POUND

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9 June 2008
SUPERSEDING
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DETAIL SPECIFICATION

TRANSMITTER, PRESSURE, OIL, SYNCHRO, 0-200 PSI 320-DEGREE MOVEMENT, 26V, SINGLE PHASE, 400 CYCLES

Inactive for new design after 28 September 1999.

This specification is approved for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of remote indicating, 320-degree movement, synchro style, differential, oil pressure transmitter.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of the documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Comments, suggestions, or questions on this document should be addressed to Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5616 or e-mailed to STDZMGT@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST data base at http://assist.daps.dla.mil.

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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.3).

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-I-7087	- Indicator, Pressure, Oil, 0-200 PSI.
MS28005	- Transmitters, Pressure, Synchro, Aircraft.
MS33678	- Connector, Receptacle, Electrical, Integral Mounting.

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	- Identification Marking of U.S. Military Property.
MIL-STD-461	- Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-1916	- DOD Preferred Methods for Acceptance of Product.

DEPARTMENT OF DEFENSE HANDBOOKS

MIL-HDBK-781	- Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.
MIL-HDBK-831	- Preparation of Test Reports.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.).

2.3 Non-government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract (see 6.3).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.100	- Engineering Drawing Practices.

(Application for copies should be addressed to the American Society of Mechanical Engineers, Three Park Avenue, New York, NY 10016-5990. Electronic copies may be obtained from http://www.asme.org/.)

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ASTM INTERNATIONAL

ASTM B 117 - Standard Practice for Operating Salt Spray (Fog) Apparatus.

(Application for copies should be addressed to the ASTM International, 100 Barr Harbor Drive, P.O. Box C700, West Conshohocken, PA 19428-2959. Electronic copies may be obtained from http://www.astm.org/.)

RADIO TECHNICAL COMMISSION FOR AERONAUTICS (RTCA)

RTCA DO-160 - Environmental Conditions and Test Procedures for Airborne Equipment.

(Application for copies should be addressed to the Radio Technical Commission for Aeronautics, Inc., 1828 L Street, NW, Suite 805, Washington, DC 20036. Electronic copies may be obtained from http://www.rtca.org/.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE AS20708 - Synchros, General Specification for.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096-0001. Electronic copies may be obtained from http://www.sae.org/.)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The transmitters furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable qualified products list (QPL) at the time set for opening of bids (see 4.3 and 6.3).

3.2 Design and construction. The transmitter shall be designed for use with a pressure indicator conforming to MIL-I-7087. The dimensional requirements shall conform to MS28005-3 except the 2.625 ± 0.005 inch bolt circle may be 2.625 ± 0.016 inch. The transmitter shall be constructed so that no parts will work loose due to strains, jars, vibrations, and other conditions incident to shipment, storage, installation, and service. Pivots, bearings, and gears shall neither bind nor shake and shall be as nearly frictionless as possible.

3.2.1 Materials. Materials used in the construction of the transmitter shall produce no toxic or deleterious fumes and shall be resistant to corrosion due to electrolytic decomposition, fungus, fuels, salt fog, and any other atmospheric condition that may be encountered during operational use or

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storage. Nonmagnetic materials shall be used except where magnetic materials are deemed essential. Cadmium shall not be used on or in the transmitter without written approval of the procuring activity.

3.2.1.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Hermetically sealed case. The case shall provide a hermetically sealed enclosure for the entire internal mechanism of the transmitter. The design of the case shall permit removal and replacement of the internal mechanism and resealing of the case. The hermetic sealing shall not be affected by any atmospheric conditions to which the transmitter may be subjected.

3.2.3 Adjustments. No special tools shall be required to adjust the transmitter. The means of adjustment shall be accessible from the exterior of the case and shall provide a minimum of 14 angular degrees of adjustment of the synchro rotor.

3.2.4 Attitude. The transmitter shall be capable of operating in any position.

3.2.5 Lubrication. The transmitter shall not require periodic lubrication.

3.2.6 Overpressure stop. An overpressure stop shall be provided and shall be designed to restrain the prime mover. The stop shall allow the indicator pointer to travel beyond full scale to a point approximately midway between the last and zero graduation.

3.2.7 Restrictors. No restrictors or capillary line connecting the pressure fitting to the pressure element shall be used that will introduce an objectionable lag in the indication response.

3.2.8 Bosses. Bosses shall be provided in accordance with MS28005 for external pressure and vent connections to the pressure sensitive element and the opposing pressure element, respectively. The bosses shall be identified with white lettering "P" and "V" permanently inscribed 0.125 inch in height.

3.2.9 Electrical connector. A hermetically sealed MS33678R14S-2P electrical connector shall be provided.

3.2.10 Weight. The weight of the transmitter shall not exceed 1.4 pounds.

3.3 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The item identification and part number requirements of ASME Y14.100 shall govern the manufacturer's part numbers and changes thereto.

3.4 Nameplate. A nameplate shall be attached to the mounting bezel with screws. Where applicable, the information marked thereon shall conform to MIL-STD-130.

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3.5 Workmanship. The transmitter shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to neatness and thoroughness of soldering, wiring, marking of parts and assemblies, plating, painting, riveting, machine screw assembly, welding, brazing, and freedom of parts from burrs and sharp edges.

3.6 Performance. The transmitter shall meet the performance requirements specified in 3.6.1 through 3.6.10 and operate within the tolerances specified in tables I and II for scale error, friction error, position error, and pointer error.

TABLE I. Scale error tolerances (± psi).

Test pressure (psi)	Room temperature	Vibration, overpressure	Low temperature, high temperature, seasoning endurance, and altitude tests
0	3.0	4.0	5.0
50	3.0	4.0	5.0
100	4.0	5.0	6.0
150	4.0	5.0	6.0
200	4.0	5.0	6.0

TABLE II. Friction, position, and pointer error tolerances (± psi).

Test	Tolerance ± psi
Friction	
Room temperature	3.0
High temperature	5.0
Low temperature	5.0
Position error at 100 psi	1.0
Vibration	
Pointer oscillation	3.0

3.6.1 Case leakage. The case leakage of the transmitter shall be no greater than 0.10 micron cubic foot per hour when the filling medium is nitrogen and helium, and no greater than 1.0 micron cubic foot per hour when the medium is 100 percent helium.

3.6.2 Electromagnetic interference. The transmitter shall meet the electromagnetic interference requirements of MIL-STD-461 for equipment and subsystems installed in, on, or launched from Air Force aircraft.

3.6.3 Overpressure. The transmitter shall withstand a pressure of 400 psi for a period of 10 minutes.

3.6.4 Indicator operation. The transmitter shall be capable of operating two MIL-I-7087 indicators without a change in performance.

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3.6.5 Seasoning. The transmitter shall withstand 10,000 applications of pressure as required to produce indications from 0 to 85 ± 5 percent of full scale and return to 0 at a rate of 30 ± 10 cycles per minute (cpm).

3.6.6 Damping. When 200 psi is applied to the transmitter, the time required for the pressure to drop to 20 psi when pressure is removed shall be 9 ± 3 seconds.

3.6.7 Torque. The transmitter shall develop a torque in the MIL-I-7087 indicator of no less than 2,000 milligram millimeters per degree of angular displacement.

3.6.8 Electrical characteristics. The transmitter synchro shall have electrical characteristics compatible with SAE AS20708 and the requirements specified in 3.6.8.1 through 3.6.8.6.

3.6.8.1 Open secondary. The current drawn by the synchro shall not exceed 240 milliamperes with secondary (stator) open-circuited.

3.6.8.2 Power consumption. The transmitter power consumption shall not exceed 6.24 volt-amperes.

3.6.8.3 Secondary voltage. The open-circuit voltage measured between any two secondary leads shall be 11.8 ± 0.3 volts alternating current (Vac). The variation between the three maximum secondary voltages on any one synchro shall not exceed 0.2 Vac.

3.6.8.4 Power supply variation. The transmitter shall operate with a supply voltage between 22 and 30Vac at a frequency between 320 and 400 hertz (Hz).

3.6.8.5 Dielectric strength. The transmitter shall withstand a potential of 250 Vac, 60 Hz for 5 seconds between each pin of the electrical connector and the transmitter case.

3.6.8.6 Electrical zero. With R2 connected to S2, and 26 Vac, 400 Hz electrical power applied to R1 and R2, the transmitter shall meet the following requirements:

a. The voltage between S1 and S3 (S3S1) shall be at minimum, in-phase with rotor voltage E (R2R1), and remain approximately in-phase as the rotor is turned counterclockwise through an angle not exceeding 180 degrees from electrical zero.

b. The voltage between R1 and S3 shall be at a minimum null and shall reach the minimum when the rotor is turned clockwise 30 degrees from electrical zero.

c. Clockwise rotation of the rotor, not exceeding 30 degrees from electrical zero, shall cause an increase in voltage between S3 and S2 and a decrease in voltage between S2 and S1.

3.6.9 Environmental conditions. The transmitter shall withstand the environmental conditions specified in 3.6.9.1 through 3.6.9.5.

3.6.9.1 Humidity. The transmitter shall withstand humidity of 95 percent.

3.6.9.2 Shock. The transmitter shall withstand a 20g shock for 11 milliseconds.

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3.6.9.3 Pressure altitude. The transmitter shall withstand a pressure altitude up to 0.64 ± 0.1 inch Hg (85,000 ± 3,000 feet).

3.6.9.4 Temperature. The transmitter shall withstand operating temperatures ranging from -67 °F to 250 °F ± 3 °F and non-operating temperatures ranging from -85 °F to 293 °F ± 3 °F.

3.6.9.5 Vibration. The transmitter shall withstand the sinusoidal vibration conditions specified in table III.

Table III. Vibration conditions.

Frequency (Hz)	Displacement in double amplitude (inch) and acceleration (g), if applicable
5-14	0.1
14-23	0.1 to 0.036 at ±1g
23-74	0.036
74-2000	0.036 to 0.01 at ±10g

3.6.10 Reliability. The transmitter shall have a minimum acceptable meantime between failures (MTBF) of 2,000 hours based on a discrimination ratio of 1.5 and a confidence level of 90 percent.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3).
b. Conformance inspection (see 4.5).

4.2 Inspection conditions. Unless otherwise specified (see 6.3), all inspections shall be performed in accordance with the conditions specified in 4.2.1 through 4.2.4.

4.2.1 Standard atmospheric conditions. Unless otherwise specified in the individual tests or the sampling tests, all tests shall be performed at ambient pressure and temperature.

4.2.2 Tapping. Unless otherwise specified in the individual tests or the sampling tests, the transmitter shall be lightly tapped before testing to overcome friction.

4.2.3 Test voltage. Unless otherwise specified in the individual tests or the sampling tests, the transmitter shall be tested with 26-V 400-Hz single-phase power. The voltage and frequency shall be maintained within ±1 percent of the nominal value.

4.2.4 Standard production indicating unit. The transmitter shall be tested with a standard production indicating unit with a maximum error of 0.5 degree.

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4.3 Qualification inspection. When qualification inspection is required (see 6.3), the test samples shall consist of 10 transmitters representative of the production process. Three of the samples shall undergo all of the tests specified in section 4.5 and seven shall be subjected to the reliability test in section 4.6.23 only.

4.4 Retention of qualification. To retain qualification, the contractor shall report at two-year intervals to the qualifying activity. The qualifying activity will establish the initial reporting date. The report shall consist of:

a. The number of lots (with total quantity identified) that have been produced during the two-year period for each QPL listed item.

b. A summary of the results of the tests performed for conformance inspection, including as a minimum, the number of lots that have passed, the number that have failed, and the cause of the failures. The test results for all reworked lots shall be identified as having passed or failed.

c. A summary of any product quality deficiency report (PQDR) submitted by a government buying activity during the two-year period and the results of those PQDRs.

d. A summary of any changes to design, manufacturing processes, materials, and subcontractors for which approval by the qualifying activity has not been obtained.

4.4.1 Reporting during periods of no production. In the event that no production occurred during the reporting period, a report shall be submitted certifying that the company still has the capabilities and facilities necessary to produce the item. If during two consecutive reporting periods, there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit the qualified products for retesting in accordance with applicable specification.

4.4.2 Failure to submit report. Failure to submit the report within 30 days after the end of each two-year period may result in loss of qualification.

4.4 Requirements cross-reference matrix. Table IV provides a cross-reference matrix of section 3 requirements to section 4 verification testing.

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Table IV. Requirement-verification cross reference matrix.

Requirement paragraph	Requirement	Verification test paragraph
3.6	Scale error tolerance	4.6.2
3.6	Friction error tolerance	4.6.3
3.6	Position error tolerance	4.6.4
3.6.1	Case leakage	4.6.5
3.6.8	Electrical characteristics	4.6.6
3.6.8.1	Open secondary	4.6.6.1
3.6.8.2	Power consumption	4.6.6.2
3.6.8.3	Secondary voltage	4.6.6.3
3.6.8.4	Power supply variation	4.6.6.4
3.6.8.5	Dielectric strength	4.6.6.5
3.6.8.6	Electrical zero	4.6.6.6
3.6.9.4	Low temperature (operational)	4.6.7
3.6.9.4	High temperature (operational)	4.6.8
3.6.9.5	Vibration	4.6.9
3.6.4	Indicator operation	4.6.10
3.6.7	Torque	4.6.11
3.6.9.4	Low temperature (non-operational)	4.6.12
3.6.9.4	High temperature (non-operational)	4.6.13
3.6.9.3	Pressure-altitude	4.6.14
3.6.2	Electromagnetic interference	4.6.15
3.6.3	Overpressure	4.6.16
3.2.1	Salt fog	4.6.1, 4.6.17
3.2.1	Fungus	4.6.1, 4.6.22
3.6.6	Damping	4.6.18
3.6.5	Seasoning	4.6.19
3.6.9.2	Shock	4.6.20
3.6.9.1	Humidity	4.6.21
3.6.10	Reliability	4.6.23

4.5 Conformance inspection. Conformance inspection shall be in accordance with MIL-STD-1916 and shall consist of the individual tests (see 4.5.1) for each unit produced as well as sampling plan A tests (see 4.5.2) or sampling plan B tests (see 4.5.3) if required by the procuring activity (see 6.3).

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4.5.1 Individual tests. Each transmitter shall be subjected to the following:

a.	Examination of product.	(4.6.1)
b.	Scale error.	(4.6.2)
c.	Friction error.	(4.6.3)
d.	Position error (attitude).	(4.6.4)
e.	Case leakage.	(4.6.5)
f.	Dielectric strength.	(4.6.6.5)

4.5.2 Sampling plan A tests. Sampling plan A tests are in addition to the individual tests. The manufacturer shall select the type of sampling plan (i.e. attribute, variable, or continuous) in accordance with MIL-STD-1916. The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

a.	Individual tests.	(4.5.1)
b.	Electrical characteristics (as listed).	
	Open secondary.	(4.6.6.1)
	Power consumption.	(4.6.6.2)
	Secondary voltage.	(4.6.6.3)
	Power supply variation.	(4.6.6.4)
c.	Low temperature (operational).	(4.6.7)
d.	High temperature (operational).	(4.6.8)
e.	Vibration.	(4.6.9)
f.	Indicator operation.	(4.6.10)

4.5.3 Sampling plan B tests. Sampling plan B tests are in addition to the individual and sampling plan A tests. The type of sampling plan (i.e. attribute, variable, or continuous) shall be selected by the manufacturer in accordance with MIL-STD-1916. The sample size shall be selected in accordance with verification level I of MIL-STD-1916.

a.	Sampling plan A tests.	(4.5.2)
b.	Electrical zero.	(4.6.6.6)
c.	Torque.	(4.6.11)
d.	Low temperature (non-operational).	(4.6.12)
e.	High temperature (non-operational).	(4.6.13)
f.	Pressure-altitude.	(4.6.14)
g.	Electromagnetic interference.	(4.6.15)
h.	Overpressure.	(4.6.16)
i.	Salt fog	(4.6.17)
j.	Damping.	(4.6.18)
k.	Seasoning.	(4.6.19)
l.	Shock.	(4.6.20)
m.	Humidity.	(4.6.21)
n.	Fungus.	(4.6.22)

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4.5.4 Rejection and retest. Any transmitter failing to meet the individual tests and the sampling plan A or B tests shall be cause for rejection of the lot. Transmitters that have been rejected may be replaced or repaired and the lot retested using a tightened sample size in accordance with verification level II of MIL-STD-1916. Prior to retesting, the manufacturer shall provide in writing to the procuring activity all corrective actions taken to correct the defect(s). Units rejected after retest shall not be resubmitted and the lot rejected unless specific approval is authorized by the procuring activity granting additional corrective actions and testing.

4.6 Test methods.

4.6.1 Examination of product. The transmitter shall be inspected to determine compliance with the requirements specified herein with respect to materials, workmanship, marking, and all other requirements not covered by tests.

4.6.2 Scale error. The transmitter shall be tested for scale errors at the test points specified in table I for both upscale and downscale readings. The test points on the upscale readings shall be measured by bringing the input reading up to, but not exceeding, the specified test point. The transmitter shall then be vibrated at a frequency of 3,600 Hz with amplitude of 0.002 inch. The difference in the scale error after vibration from the true input value shall not exceed the scale error tolerance specified in table I. The downscale error shall be measured in the same manner, except that the input reading value shall be brought down to, but not below, the specified test point.

4.6.3 Friction error. The difference in transmitter output prior to vibration specified in 4.6.2 and after application of this vibration is friction error. Friction error shall not exceed the tolerance specified in table II.

4.6.4 Position error (attitude). The transmitter shall be energized to obtain a reading of one half scale on the indicator conforming to MIL-I-7087. The change in indication produced by rotating the transmitter from the normal operating position to a position 90 degrees clockwise and counterclockwise (longitudinal axis vertical) shall not exceed the tolerance specified for position error in table II. The position error test may be conducted in conjunction with the scale error test.

4.6.5 Case leakage. The transmitter case shall be tested for leaks by means of a mass spectrometer helium detector. Where a nitrogen-helium mixture is used as the fill medium, the detected leak rate shall not exceed 0.10 micron cubic foot per hour. Where 100 percent helium is used as the fill medium, the leak rate shall not exceed 1.0 micron cubic foot per hour.

4.6.6 Electrical characteristics. The transmitter shall be tested in accordance with 4.6.6.1 through 4.6.6.6 to determine compliance with 3.6.8.

4.6.6.1 Open secondary. The transmitter shall be tested for compliance with 3.6.8.1.

4.6.6.2 Power consumption. The transmitter shall be tested for compliance with 3.6.8.2.

4.6.6.3 Secondary voltage. The transmitter shall be tested for compliance with 3.6.8.3.

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4.6.6.4 Power supply variation. The transmitter shall be subjected to the scale error and friction error tests during which the voltage and frequency shall be varied from 22V to 30V and 320 Hz to 400 Hz (or any combination thereof). The scale and friction errors shall not exceed those specified in tables I and II.

4.6.6.5 Dielectric strength. A potential of 250 Vac shall be applied between each pin of the electrical connector and the case for 5 seconds. There shall be no insulation breakdown.

4.6.6.6 Electrical zero. The transmitter shall have R2 connected to S2, and 26 Vac, 400 Hz shall be applied to R1 and R2 for the following tests.

a. The electrical zero shall be found. It shall be where the voltage between S1 and S3 is at a minimum.

b. The rotor shall be set to electrical zero. The rotor shall be rotated counterclockwise 180 degrees. The voltage S3S1 shall remain approximately in phase with the rotor voltage between R1 and R2 throughout the entire 180 degree range.

c. The rotor shall be set to electrical zero and the voltage shall be near the minimum. The rotor shall be rotated clockwise. The voltage between R1 and S3 shall be at a minimum approximately 30 degrees clockwise of the electrical zero.

d. The rotor shall be returned to electrical zero. The rotor shall be rotated 30 degrees clockwise from electrical zero. The voltage between S3 and S2 shall increase and the voltage between S2 and S1 shall decrease during the rotation.

4.6.7 Low temperature (operational). The transmitter shall be subjected to a temperature of -67 °F ± 3 °F for 4 hours while electrically connected to the MIL-I-7087 indicator with power applied. At the end of the 4-hour period and while still at the specified temperature, the transmitter shall be subjected to the scale error and friction error tests. The errors shall not exceed the tolerances specified in tables I and II.

4.6.8 High temperature (operational). The transmitter shall be subjected to a temperature of 250 °F ± 3 ºF for 6 hours while electrically connected to the MIL-I-7087 indicator with power applied. At the end of the 6-hour period and while still at the specified temperature, the transmitter shall be subjected to the scale error and friction error tests. The tolerances shall not exceed those specified in tables I and II.

4.6.9 Vibration. The transmitter shall be subjected to a vibration test in accordance with RTCA DO-160, using test levels specified in table III herein. At the end of this test, the transmitter shall be subjected to the scale error test for upscale readings only. The scale errors shall not exceed the tolerances specified in table I.

4.6.10 Indicator operation. The transmitter shall be subjected to scale error tests while electrically connected to one MIL-I-7087 indicator. The transmitter shall then be subjected to scale error tests while electrically connected to two MIL-I-7087 indicators. There shall be no change in indicator readings.

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4.6.11 Torque. The torque requirement shall be verified by a test designed by the manufacturer.

4.6.12 Low temperature (non-operational). The transmitter shall be subjected to a temperature of -85 °F ± 3 ºF for 50 hours. The temperature shall then be raised to -65 °F ± 3 ºF for 4 hours. The transmitter shall then be returned to room temperature for 4 hours and then subjected to the scale error and friction error tests. The tolerances shall not exceed those specified in tables I and II.

4.6.13 High temperature (non-operational). The transmitter shall be subjected to a temperature of 293 °F ± 18 ºF for 80 hours. The transmitter shall then be returned to room temperature for 4 hours and then subjected to the scale error and friction error tests. The tolerances shall not exceed those specified in tables I and II.

4.6.14 Pressure-altitude. The transmitter shall be subjected to a pressure altitude test with an absolute internal chamber pressure of 0.64 ± 0.1 inch Hg (85,000 ± 3,000 feet). The transmitter vent shall be held at the prevailing atmospheric pressure. The transmitter shall then be subjected to the scale error test for upscale pressure only.

4.6.15 Electromagnetic interference. The transmitter shall be subjected to and shall meet the electromagnetic interference tests CE102, CS101, CS114, CS115, CS116, RE102, and RS103 of MIL-STD-461 for Air Force aircraft.

4.6.16 Overpressure. The transmitter shall be subjected to a pressure of 400 psi for a period of 10 minutes. The transmitter shall be tested not less than 1 hour following the overpressure test for scale error as specified under individual tests on increasing pressure only. The scale error shall not exceed the tolerance specified in table I.

4.6.17 Salt fog. The transmitter shall be subjected to the humidity test in ASTM B 117. The transmitter shall be examined, and there shall be no evidence of external corrosion or deterioration. The transmitter shall then be subjected to the dielectric strength test. The impedance between the case and each pin shall be measured and shall be a minimum of 20 megohms. The external surface of the transmitter may be washed with distilled water and air dried prior to its subjection to the dielectric strength test.

4.6.18 Damping. A pressure sufficient to produce full-scale pointer deflection shall be applied to the transmitter. A quick release valve shall be used to exhaust the pressure. The time required for the pointer to traverse the interval from 200 psi to 20 psi shall be 9 ± 3 seconds. The test shall be conducted with lubricating oil with a viscosity of approximately 2,500 centistokes, such as grade 1005 oil at -65 ºF, or equivalent.

4.6.19 Seasoning. The transmitter shall be subjected to 10,000 pressure cycles that produce a movement of the MIL-I-7087 indicator from 0 to 85 ± 5 percent of full scale and return to 0 at a rate of 30 ± 10 cpm. One hour after completion of this test, the transmitter shall be subjected to the scale error test at room temperature and shall meet the tolerances specified in tables I and II.

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4.6.20 Shock. The transmitter shall be subjected to a shock of 20g for 11 milliseconds while electrically connected to the MIL-I-7087 indicator with power applied. The transmitter shall be subjected to the scale error and friction error tests. The tolerances shall not exceed those specified in tables I and II.

4.6.21 Humidity. The transmitter shall be subjected to the humidity test in accordance with category B of RTCA DO-160 while electrically connected to the MIL-I-7087 indicator with power applied. Immediately upon completion of the test, the transmitter shall be externally dried and subjected to scale error, friction error, and dielectric strength. The tolerances shall not exceed those specified in section 3.

4.6.22 Fungus. The transmitter shall be subjected to the fungus test in RTCA DO-160.

4.6.23 Reliability test plan and report. Prior to starting the reliability test, the manufacturer shall submit a detailed reliability test plan for formal approval by the procuring activity. The manufacturer may refer to the requirements specified in Test Plan I-D of MIL-HDBK-781 as a guide for the reliability test plan. The manufacturer shall prepare a final report that summarizes the achieved reliability of the test samples. The report shall also contain test procedures, details of any malfunctions, and applicable charts and graphs showing failures/MTBF plotted against test time and other pertinent test data. The manufacturer may refer to the guidelines outlined in MIL-HDBK-831 for preparing the report.

4.6.23.1 Reliability test. The seven transmitters shall each accumulate 2,000 hours of test time by being subjected to a series of environmental test cycles. The transmitters shall be checked every 25 test cycles (250 hours) for scale error. The scale error shall not exceed 150 percent of the room temperature tolerances specified in table I. Each test cycle is approximately 10 hours in duration and shall meet the following requirements:

a. The test cycle shall begin when rated power (26 ± 2 Vac at 400 ± 5 Hz) is applied to the transmitter and the temperature is at -67 °F. The temperature shall be cycled as follows: 4 hours at -67 °F, increase temperature to 250 °F at a rate of 9 °F per minute; 4 hours at 250 °F, lower temperature to -67 °F at a rate of 9 °F per minute, and repeat cycle. Prior to the temperature changing from low to high and high to low the power shall be turned off for a period of 10 minutes during each 10-hour cycle.

b. Oil pressure shall be applied to the transmitter at a rate of 3 to 4 cpm from 0 to 85 ± 5 percent of full scale and return to 0 for the duration of the 2,000 hours of testing.

c. The transmitter shall be vibrated in accordance with 4.6.9 for 10 minutes out of every hour of power "ON" time.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.3). When packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The oil pressure transmitter covered by this specification is intended to transmit the oil pressure of the lubricating system of aircraft engines to an applicable indicator.

6.2 Qualified products. With respect to products requiring qualification, awards will be made only for products that are, at the time of award of contract, qualified for inclusion in QPL-5790 whether or not such products have actually been so listed by that date. The attention of contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Defense Supply Center Richmond, ATTN: DSCR-VEB, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

6.3 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. The specific issue of individual documents referenced (see 2.1 and 2.3).
c. Special test conditions, if any (see 4.2).
d. When qualification inspection is required (see 4.3).
e. Required sampling plan tests A or B (see 4.5).
f. Required level of packaging and packing (see 5.1).

6.4 Subject term (key word) listing.

aircraft
differential
lubricating
remote indicating

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6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notions. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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